

AMAN SHUKLA

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PROFESSIONAL SUMMARY

AI/ML Scientist with 5+ years bridging SOTA research and production systems across multimodal architectures and transformer-based representation learning. Experienced in the full research stack from data pipelines to models to scalable distributed training, post-training, mechanistic interpretability and deployment on AWS with work shipping into systems driving \$1M+ in pipeline. Published at NeurIPS, ICLR, and ICML, with a track record of turning ambiguous research problems into deployed, evaluated products.

WORK EXPERIENCE

AI/ML Data Scientist, Resonate, Remote, USA Oct 2023-Present

- Led **end-to-end research-to-production** development of AMA engine; from prototype to deployment on cloud.
- Evaluated **multi-agent activation** workflows in the SaaS platform using **LLM-as-a-judge** evaluation framework.
- Built a **mechanistic interpretability** framework using **SAEs** to generate monosemantic embeddings providing clients with explainable, responsible AI insights.
- Architected **MoE** with data-parallel distributed training across **multi-GPU (P2P, NCCL) on AWS** resolving bottlenecks
- Researched optimal **geospatial** representations for transformer-based attribute prediction learning to encode location signals into latent embeddings that surface unobserved attributes
- Initiated & led **Embeddings** product, generating ~\$1M in pipeline, including client presentations, **technical sales** support, and privacy solutions.
- Spearheaded vendor **A/B testing** and mentored 2 engineers to develop 10+ models across 6 vendors in 3 weeks, drastically improving evaluation to production cycle.
- Contributed to AI research by authoring papers at top conferences including **ICML, NeurIPS and IEEE ICDM**, focusing on personalization and behavior in advertisement technology.

ML Research Engineer, MusicXLab (NYU), Remote, USA Jan 2022-Oct 2023

- Developed a **multimodal** generative system conditioning **MusicGen** on **emoji-derived** semantic embeddings mapping discrete visual tokens to continuous latent representations to steer audio generation toward target emotions.
- Enabled emotion-driven **music generation** by prompts, allowing personalized audio tailored to the user's intended emotion.
- Fine-tuned DeepMoji on DALI using **SFT** to build an emoji-to-emotion retrieval model; outputs served as conditioning signals bridging visual/symbolic input to generative audio output.

Data Scientist, Capgemini, Bengaluru, India Jul 2018-Dec 2020

- Engineered extraction tools for map features(bridges,tunnels) using **pandas** and **numpy** to support autonomous driving.
- Created framework for NDS HD map extraction achieving a **60% reduction** in **PostgreSQL** query latency.
- Integrated coordinate mapping for ECU decision-making using **orthogonal mapping(cosine similarity)** and **shapely** ensuring high-precision spatial positioning.

PUBLICATIONS

Studied the effect of VIB compression on RLHF safety alignment via SFT using mechanistic interpretation ([Under Review, EMNLP 26](#))

Responsible Imputations of User Behavior Surveys via Mask-Aware Transformers [Paper](#)

Reliable ML@**NeurIPS 2025**

Towards Fair User Modeling from Survey-Guided Learning ([Under Proceedings](#))

IEEE FMLDS 2025

Modeling User Behavior from Adaptive Surveys with Supplemental Context (Best Paper) [Paper](#)

NewInML@ **ICML 2025**

A Tale of Two Circuits: Grokking as Competition of Sparse and Dense Subnetworks [Paper](#), [GitHub](#)

ME-FoMo@ **ICLR 2023**

EmoMusic: New Fun and Interactive Way to Listen to Music [Paper](#), [GitHub](#), [Talk](#)

MILC@ **IUI 2023**

EDUCATION

New York University, Courant Institute of Mathematics & Science Master of Science, Computer Science

National Institute of Technology Hamirpur, Bachelor of Technology, Electronics and Communication Engineering

TECHNICAL SKILLS

Programming: Python, C++, C, PostgreSQL, SQL, REST API, Linux, Scala, CUDA

Frameworks: Pytorch, Jupyter, TensorFlow, HuggingFace, Spark, Git, AWS, Keras, Slurm, Docker, Kubernetes